

CS430 Final Project: Spending Tracker

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Introduction

Purpose

Our goal when creating this project was to make a spending tracking tool we would actually use. This web app is a place where purchases can be stored, recalled, and analyzed. We didn't like modern budgeting apps with the extra fluff of telling you where and how your money should be spent. The idea of something smaller and specialized created this spending tracker, where the only goal is to store where your money is spent. We track what you spent money on categorically, as well as the amount, the location, the time and date, the medium of payment, and whether or not it was spent online. This allows for analytics to be generated that show your spending habits in a digestible format.

Scope

The scope of this project is intentionally narrow. We did not intend to compete with bigger budgeting apps who have more features and better analytics. Our sole goal was to create a centralized ledger and analysis tool to show you where your money ends up. This required the creation of one tool, the web app itself. Included in this are portions to log in, view enter new purchases, and look at analytics for past purchases.

Details

This project has a relatively straightforward 'how'. It is a server hosted web application, hosted in our basement on Ubuntu server. The app uses flask for the web framework to keep everything simple and light. The backend database is hosted on Microsoft SQL Server 2025. Plotly was used for the creation of all the graphs and charts. Implementation of the login system is done with cookies, as modern pages do.

Contributions

Cameron

- Frontend
 - HTML
 - CSS
 - JS
 - Password authentication system
 - Flask implementation
 - Forms for data collection
 - Graphical design
 - CrewAI Integration

Jack

- Backend
 - Database schema
 - Database setup
 - Fields engineering/ERD generation
 - SQL queries for sending data to frontend
 - Graph creation with plotly
 - Creation of synthetic data

Usage

Launch

This is a server hosted web app so the only requirements to launch are a modern browser. The user simply navigates to the address and the app is presented.

Features

- Secure log in system with cookies
- Password changing
- Add new purchases
- See previous 5 purchases
- See spending over time as a stacked area chart

- See your purchases by subcategory as a percentage of total purchases and amount spent
- See where you spend your money with a map chart

Technical Breakdown

Component Implementation

Packages/Tools